Colligative properties

1. **(a) Lower**

   Explanation:

   Colligative properties depend on the **number of particles** present in the solution. In colloidal dispersions, particles are very large, so the number of particles becomes less. Therefore, colloidal dispersions show lower colligative properties than true solutions.

2. **(d) Different boiling and different freezing points**

   Explanation:

   Boiling point elevation and freezing point depression are **colligative properties**. They depend on:

   - number of solute particles
   - van't Hoff factor (i)

   Even if two solutions are equimolar, different solutes may produce different numbers of particles in solution due to dissociation or association.

   Therefore, their boiling points and freezing points can be different.

3. (a) Osmotic pressure is colligative property.

4. **(c) Number of solute particles present in it**

   Explanation:

   Colligative properties depend only on the **number of solute particles** present in the solution and not on the nature of the solute. Examples include:

   - lowering of vapour pressure
   - elevation of boiling point
   - depression of freezing point
   - osmotic pressure.

5. (c) Vapour pressure is not colligative property.

6. **(a) Optical activity**

   Explanation:

Optical activity depends on the nature of the substance, not on the number of particles. Therefore, it is not a colligative property.

7. **(b) The relative number of solute and solvent particles**

Explanation:

Colligative properties depend only on the number of solute particles relative to solvent particles.

8. **(a) Refractive index**

Explanation:

Refractive index depends on the nature of the substance, so it is not a colligative property.

9. **(c) Osmotic pressure**

Explanation:

Osmotic pressure depends on the number of solute particles present in the solution.

10. **(a) Molar Mass**

Explanation:

Colligative properties help determine the molar mass of solutes because they depend on the number of particles in solution.

11. **(c) Molality**

Explanation:

Molality depends on:

- moles of solute
- mass of solvent

Since mass does not change with temperature, molality remains constant.